Deep Generative Models

Lecture 6

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2026, Autumn

Recap of Previous Lecture

Theorem

The minimax game

$$\min_G \max_D \Big[\underbrace{\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D(\mathbf{G}(\mathbf{z})))}_{V(G,D)} \Big]$$

achieves its global optimum precisely when $p_{\text{data}}(\mathbf{x}) = p_{\boldsymbol{\theta}}(\mathbf{x})$, and $D^*(\mathbf{x}) = 0.5$.

$$V(G, D^*) = 2\, D_{\mathrm{JS}}(p_{\text{data}}(\mathbf{x}) \,\|\, p_{\boldsymbol{\theta}}(\mathbf{x})) - 2 \log 2, \quad V(G^*, D^*) = -2 \log 2.$$

Expectations

If the generator can express **any** function and the discriminator is **optimal** at every step, the generator **will converge** to the target distribution.

Goodfellow I. J. et al. Generative Adversarial Networks, 2014

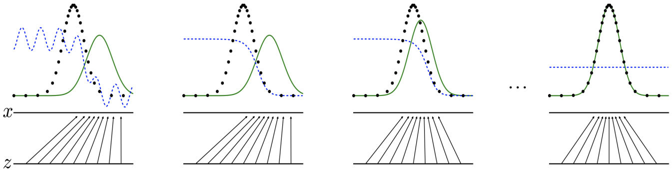
Recap of Previous Lecture

Reality

- Generator updates are performed in parameter space, and the discriminator is often imperfectly optimized.
- Generator and discriminator losses typically oscillate during GAN training.

Objective

$$\min_{\boldsymbol{\theta}} \max_{\boldsymbol{\phi}} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D_{\boldsymbol{\phi}}(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D_{\boldsymbol{\phi}}(\mathbf{G}_{\boldsymbol{\theta}}(\mathbf{z}))) \right]$$



(a) (b) (c) (d)

Mode collapse refers to the phenomenon where the generator in a <u>GAN produces only one</u> or a few different modes of the distribution.

Goodfellow I. J. et al. Generative Adversarial Networks, 2014
Metz L. et al. Unrolled Generative Adversarial Networks, 2016

Recap of Previous Lecture

Standard GAN Objective

$$\min_{\boldsymbol{\theta}} \max_{\boldsymbol{\phi}} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D_{\boldsymbol{\phi}}(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D_{\boldsymbol{\phi}}(\mathbf{G}_{\boldsymbol{\theta}}(\mathbf{z})))\right]$$

Theoretical Results

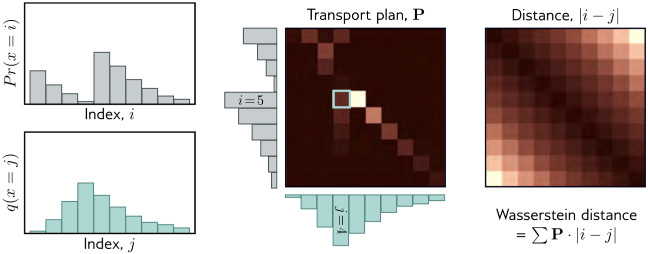
- Both the data distribution $p_{\text{data}}(\mathbf{x})$ and the generative distribution $p_{\boldsymbol{\theta}}(\mathbf{x})$ are supported on low-dimensional manifolds.
- If $p_{\text{data}}(\mathbf{x})$ and $p_{\boldsymbol{\theta}}(\mathbf{x})$ are disjoint, a smooth optimal discriminator can exist!

For such low-dimensional, disjoint manifolds:

$$D_{\mathrm{KL}}(p_{\text{data}} \| p_{\boldsymbol{\theta}}) = D_{\mathrm{KL}}(p_{\boldsymbol{\theta}} \| p_{\text{data}}) = \infty, \quad D_{\mathrm{JS}}(p_{\text{data}} \| p_{\boldsymbol{\theta}}) = \log 2$$

Weng L. From GAN to WGAN, 2019
Arjovsky M., Bottou L. Towards Principled Methods for Training Generative Adversarial Networks, 2017

Recap of Previous Lecture



Wasserstein Distance

$$W(\pi \| p) = \inf_{\gamma \in \Gamma(\pi, p)} \mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\| = \inf_{\gamma \in \Gamma(\pi, p)} \int \|\mathbf{x}_1 - \mathbf{x}_2\| \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 d\mathbf{x}_2$$

- $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ is the transport plan: the amount of "dirt" assigned from $\mathbf{x}_1$ to $\mathbf{x}_2$.
- $\Gamma(\pi, p)$ denotes the set of all joint distributions $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ with marginals π and p ($\int \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 = p(\mathbf{x}_2)$, $\int \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_2 = \pi(\mathbf{x}_1)$).
- $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ is the mass, $\|\mathbf{x}_1 - \mathbf{x}_2\|$ is the distance.

Simon J.D. Prince. Understanding Deep Learning, 2023
Arjovsky M., Chintala S., Bottou L. Wasserstein GAN, 2017

Recap of Previous Lecture

Theorem (Kantorovich-Rubinstein Duality)

$$W(p_{\text{data}} \| p_{\boldsymbol{\theta}}) = \frac{1}{K} \max_{\|f\|_L \leq K} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} f(\mathbf{x}) \right],$$

where $f : \mathbb{R}^m \to \mathbb{R}$ is K-Lipschitz continuous ($\|f\|_L \leq K$).

WGAN Objective

$$\min_{\boldsymbol{\theta}} W(p_{\text{data}} \| p_{\boldsymbol{\theta}}) \approx \min_{\boldsymbol{\theta}} \max_{\phi \in \boldsymbol{\Phi}} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f_\phi(\mathbf{x}) - \mathbb{E}_{p(\mathbf{z})} f_\phi(\mathbf{G}_{\boldsymbol{\theta}}(\mathbf{z})) \right].$$

- The discriminator D is replaced by function f: in WGAN, it is known as the **critic**, which is *not* a classifier.
- If the weights ϕ are restricted to a compact set $\boldsymbol{\Phi} = [-c, c]^d$, then $f_\phi(\mathbf{x})$ is K-Lipschitz continuous.

$$K \cdot W(p_{\text{data}} \| p_{\boldsymbol{\theta}}) = \max_{\|f\|_L \leq K} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} f(\mathbf{x}) \right] \geq$$

$$\geq \max_{\phi \in \boldsymbol{\Phi}} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f_\phi(\mathbf{x}) - \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} f_\phi(\mathbf{x}) \right]$$

Arjovsky M., Chintala S., Bottou L. Wasserstein GAN, 2017

Recap of Previous Lecture

Frechet Inception Distance (FID)

For normal distributions $p_{\text{data}}(\mathbf{x}_1) = \mathcal{N}(\boldsymbol{\mu}_{\text{data}}, \boldsymbol{\Sigma}_{\text{data}})$, $p(\mathbf{x}_2) = \mathcal{N}(\boldsymbol{\mu}_{\boldsymbol{\theta}}, \boldsymbol{\Sigma}_{\boldsymbol{\theta}})$:

$$\mathrm{FID}(p_{\text{data}}, p_{\boldsymbol{\theta}}) = W_2^2(p_{\text{data}} \| p_{\boldsymbol{\theta}}) = \inf_{\gamma \in \Gamma(p_{\text{data}}, p_{\boldsymbol{\theta}})} \mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\|^2$$
$$= \|\boldsymbol{\mu}_{\text{data}} - \boldsymbol{\mu}_{\boldsymbol{\theta}}\|^2 + \mathrm{tr}\left[\boldsymbol{\Sigma}_{\text{data}} + \boldsymbol{\Sigma}_{\boldsymbol{\theta}} - 2\left(\boldsymbol{\Sigma}_{\text{data}}^{1/2} \boldsymbol{\Sigma}_{\boldsymbol{\theta}} \boldsymbol{\Sigma}_{\text{data}}^{1/2}\right)^{1/2}\right]$$

Drawbacks

- Depends on the pretrained classification network.
- Uses the normality assumption.
- May not correlate with human evaluation.

Heusel M. et al. GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium, 2017
Jayasumana S. et al. Rethinking FID: Towards a Better Evaluation Metric for Image Generation, 2024

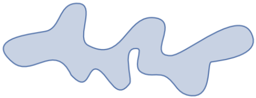
Recap of Previous Lecture

- $\mathcal{S}_{\text{data}} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\text{data}}(\mathbf{x})$ – real samples;
- $\mathcal{S}_{\boldsymbol{\theta}} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\boldsymbol{\theta}}(\mathbf{x})$ – generated samples.

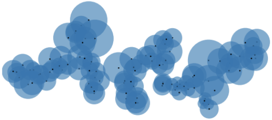
Define a binary function:

$$\mathbb{I}(\mathbf{x}, \mathcal{S}) = \begin{cases} 1, & \text{if } \exists\, \mathbf{x}' \in \mathcal{S} : \|\mathbf{x} - \mathbf{x}'\|_2 \leq \|\mathbf{x}' - \mathrm{NN}_k(\mathbf{x}', \mathcal{S})\|_2; \\ 0, & \text{otherwise.} \end{cases}$$

$$\Pr(\mathcal{S}_{\text{data}}, \mathcal{S}_{\boldsymbol{\theta}}) = \frac{1}{n} \sum_{\mathbf{x} \in \mathcal{S}_{\boldsymbol{\theta}}} \mathbb{I}(\mathbf{x}, \mathcal{S}_{\text{data}}); \quad \mathrm{Rec}(\mathcal{S}_{\text{data}}, \mathcal{S}_{\boldsymbol{\theta}}) = \frac{1}{n} \sum_{\mathbf{x} \in \mathcal{S}_{\text{data}}} \mathbb{I}(\mathbf{x}, \mathcal{S}_{\boldsymbol{\theta}}).$$



(a) True manifold



(b) Approx. manifold

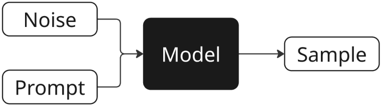
Embed the samples using a pretrained network (as in FID).

Kynkäänniemi T. et al. Improved precision and recall metric for assessing generative models, 2019

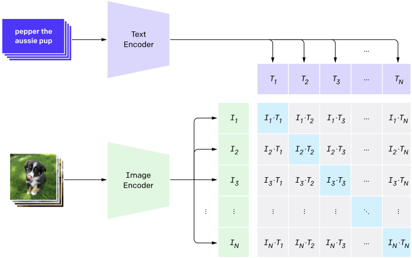
Recap of Previous Lecture

Unconditional Model

Conditional Model



We need a way to measure not only the quality of the generated image, but also how well it's aligned with the prompt.



Radford A. et al. Learning transferable visual models from natural language supervision, 2021

Recap of Previous Lecture

- No automated metric is perfect.
- The best way to evaluate generative models is by human assessment.
- It's important to assess various properties.

Аспект	Yandex ART 2.0	Mj 6.1	Mj 6	Ideogram	Recraft	Google Imagen3	Dall-E 3	FLUX	SBER Kandi3.1
Релевантность	0,59	0,58	0,63	0,45	0,51	0,50	0,50	0,54	0,75
Эстетика	0,49	0,55	0,55	0,51	0,51	0,61	0,61	0,54	0,59
Комплексность	0,44	0,73	0,70	0,68	0,76	0,75	0,75	0,71	0,74
Дефектность	0,69	0,57	0,68	0,55	0,59	0,63	0,63	0,50	0,75
Предпочтение	0,66	0,60	0,69	0,49	0,54	0,63	0,63	0,51	0,84

YandexART 2.5, 2025

Outline

1. Langevin Dynamics

2. Score Matching
   - Denoising Score Matching
   - Noise-Conditioned Score Network (NCSN)

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Energy-Based Models

Unnormalized Density

$$p_{\boldsymbol{\theta}}(\mathbf{x}) = \frac{\hat{p}_{\boldsymbol{\theta}}(\mathbf{x})}{Z_{\boldsymbol{\theta}}}, \quad \text{where } Z_{\boldsymbol{\theta}} = \int \hat{p}_{\boldsymbol{\theta}}(\mathbf{x}) d\mathbf{x}$$

- $\hat{p}_{\boldsymbol{\theta}}(\mathbf{x})$ can be any non-negative function.
- If we reparameterize as $\hat{p}_{\boldsymbol{\theta}}(\mathbf{x}) = \exp(-f_{\boldsymbol{\theta}}(\mathbf{x}))$, we eliminate the non-negativity constraint.

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Log-Density Gradient

The gradient of the normalized log-density equals that of the unnormalized log-density:

$$\nabla_{\mathbf{x}} \log p_{\boldsymbol{\theta}}(\mathbf{x}) = \nabla_{\mathbf{x}} \log \hat{p}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log Z_{\boldsymbol{\theta}} = \nabla_{\mathbf{x}} \log \hat{p}_{\boldsymbol{\theta}}(\mathbf{x})$$

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- Suppose we already have this density (normalized or not) $p_{\boldsymbol{\theta}}(\mathbf{x})$.
- How can we sample from the model?

Langevin Dynamics

Theorem (Informal)

Let $\mathbf{x}_0$ be a random vector. Under mild regularity conditions, samples from the following dynamics will eventually follow $p_{\boldsymbol{\theta}}(\mathbf{x})$ (for sufficiently small η and large l):

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}_l} \log p_{\boldsymbol{\theta}}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l, \quad \boldsymbol{\epsilon}_l \sim \mathcal{N}(0, \mathbf{I}).$$

Lai C. H. et al. The principles of diffusion models, 2025.

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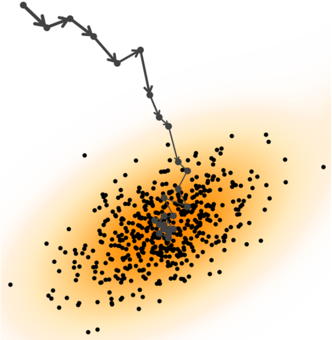
- What if $\boldsymbol{\epsilon}_l = \mathbf{0}$?
- The density $p_{\boldsymbol{\theta}}(\mathbf{x})$ is the **stationary** distribution of the Markov chain.
- The gradient is taken with respect to $\mathbf{x}$, not $\boldsymbol{\theta}$.
- $\nabla_{\mathbf{x}} \log p_{\boldsymbol{\theta}}(\mathbf{x})$ defines a vector field.



Lai C. H. et al. The principles of diffusion models, 2025.

Outline

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2. Score Matching
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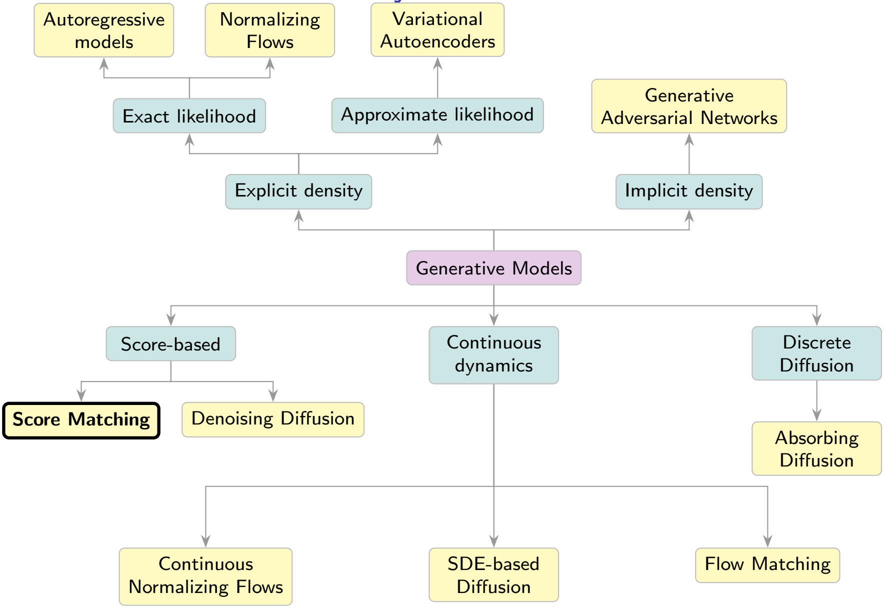
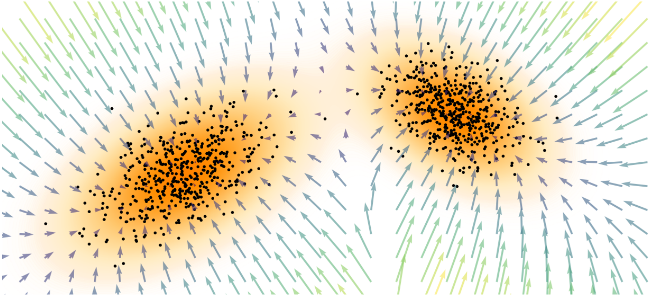
Generative Models Taxonomy



Score Function

$$\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) = \nabla_{\mathbf{x}} \log p_{\boldsymbol{\theta}}(\mathbf{x})$$



Lai C. H. et al. The principles of diffusion models, 2025.

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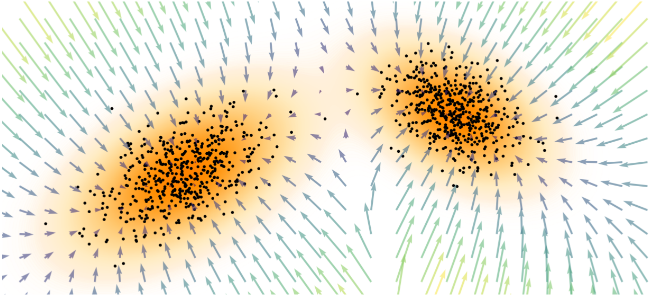


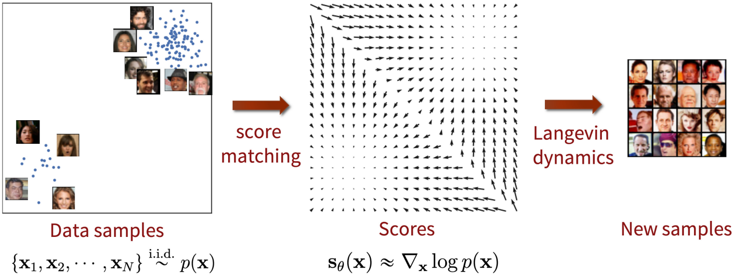
Langevin Dynamics

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}_l} \log p_{\text{data}}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l$$

$$= \mathbf{x}_l + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}_l} \log p_{\boldsymbol{\theta}}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l = \mathbf{x}_l + \frac{\eta}{2} \cdot \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l.$$

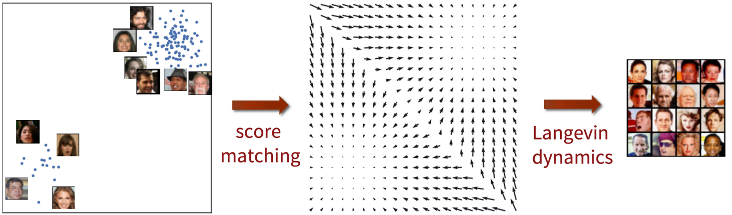
Lai C. H. et al. The principles of diffusion models, 2025.

Score Matching



Song Y. Generative Modeling by Estimating Gradients of the Data Distribution, blog post, 2021

Score Matching



Data samples $\{\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_N\} \overset{\text{i.i.d.}}{\sim} p(\mathbf{x})$

Scores $\mathbf{s}_\theta(\mathbf{x}) \approx \nabla_\mathbf{x} \log p(\mathbf{x})$

New samples

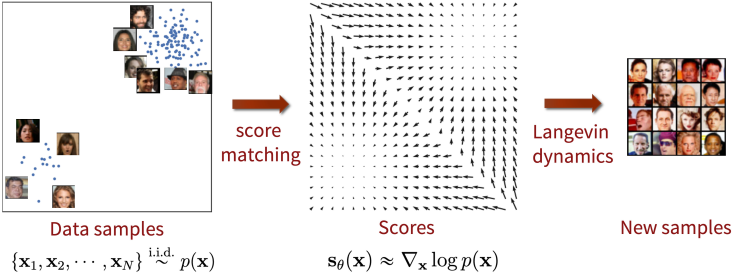
Fisher Divergence

$$D_F(p_{\text{data}}, p_{\boldsymbol{\theta}}) = \frac{1}{2}\mathbb{E}_{p_{\text{data}}} \left\| \nabla_\mathbf{x} \log p_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_\mathbf{x} \log p_{\text{data}}(\mathbf{x}) \right\|_2^2 =$$

$$= \frac{1}{2}\mathbb{E}_{p_{\text{data}}} \left\| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_\mathbf{x} \log p_{\text{data}}(\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

Song Y. Generative Modeling by Estimating Gradients of the Data Distribution, blog post, 2021

Score Matching



Fisher Divergence

$$
\begin{aligned}
D_F(p_{\text{data}}, p_{\boldsymbol{\theta}}) &= \frac{1}{2}\mathbb{E}_{p_{\text{data}}} \left\| \nabla_{\mathbf{x}} \log p_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) \right\|_2^2 = \\
&= \frac{1}{2}\mathbb{E}_{p_{\text{data}}} \left\| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}
\end{aligned}
$$

Problem: We don't know $\nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$.

Song Y. Generative Modeling by Estimating Gradients of the Data Distribution, blog post, 2021

Why Is the Score Function Better than the Density?

Freedom from Normalization Constants

Many distributions are defined only up to an unnormalized density $\hat{p}(\mathbf{x})$:

$$\nabla_{\mathbf{x}} \log p(\mathbf{x}) = \nabla_{\mathbf{x}} \log \hat{p}(\mathbf{x}) - \underbrace{\nabla_{\mathbf{x}} \log Z}_{=0} = \nabla_{\mathbf{x}} \log \hat{p}(\mathbf{x})$$

The score function bypasses the intractable constant Z entirely!

Lai C. H. et al. The principles of diffusion models, 2025.

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The score function bypasses the intractable constant Z entirely!

A Complete Representation

The score function fully characterizes the underlying distribution (up to a constant):

$$\log p(\mathbf{x}) = \log p(\mathbf{x}_0) + \int_0^1 \mathbf{s}(\mathbf{x}_0 + t(\mathbf{x} - \mathbf{x}_0))^\top (\mathbf{x} - \mathbf{x}_0) dt$$

Modeling the score is as expressive as modeling $p(\mathbf{x})$ itself, while often more tractable for generative modeling.

Lai C. H. et al. The principles of diffusion models, 2025.

Outline

1. Langevin Dynamics

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Denoising Score Matching

Let us perturb the original data $\mathbf{x} \sim p_{\text{data}}(\mathbf{x})$ with Gaussian noise:

$$\mathbf{x}_\sigma = \mathbf{x} + \sigma \cdot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}), \quad q(\mathbf{x}_\sigma | \mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma^2 \cdot \mathbf{I})$$

Vincent P. A Connection Between Score Matching and Denoising Autoencoders, 2010

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$$q(\mathbf{x}_\sigma) = \int q(\mathbf{x}_\sigma|\mathbf{x}) p_{\text{data}}(\mathbf{x}) d\mathbf{x}.$$

Assumption

The solution to

$$\frac{1}{2}\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

satisfies $\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) \approx \mathbf{s}_{\boldsymbol{\theta},0}(\mathbf{x}_0) = \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x})$ if σ is sufficiently small.

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satisfies $\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) \approx \mathbf{s}_{\boldsymbol{\theta},0}(\mathbf{x}_0) = \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x})$ if σ is sufficiently small.

- The score function of the noised data nearly matches the score function of the original data.
- The score function $\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma)$ is parameterized by σ.
- **Note:** We don't know $q(\mathbf{x}_\sigma)$, just as we don't know $p_{\text{data}}(\mathbf{x})$.

Vincent P. A Connection Between Score Matching and Denoising Autoencoders, 2010

Denoising Score Matching

Theorem

Under mild regularity conditions, the following holds:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \big\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \big\|_2^2 =$$
$$= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \big\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \big\|_2^2 + \text{const}(\boldsymbol{\theta})$$

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Theorem

Under mild regularity conditions, the following holds:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \big\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \big\|_2^2 =$$
$$= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \big\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \big\|_2^2 + \text{const}(\boldsymbol{\theta})$$

Gradient of the Noise Kernel

$$\mathbf{x}_\sigma = \mathbf{x} + \sigma \cdot \boldsymbol{\epsilon}, \quad q(\mathbf{x}_\sigma|\mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma^2 \cdot \mathbf{I})$$

Vincent P. A Connection Between Score Matching and Denoising Autoencoders, 2010

Denoising Score Matching

Theorem

Under mild regularity conditions, the following holds:

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Gradient of the Noise Kernel

$$\mathbf{x}_\sigma = \mathbf{x} + \sigma \cdot \boldsymbol{\epsilon}, \quad q(\mathbf{x}_\sigma|\mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma^2 \cdot \mathbf{I})$$

$$\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) = -\frac{\mathbf{x}_\sigma - \mathbf{x}}{\sigma^2} = -\frac{\boldsymbol{\epsilon}}{\sigma}$$

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Denoising Score Matching

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Gradient of the Noise Kernel

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- The right-hand side doesn't require computing $\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)$ or even $\nabla_{\mathbf{x}_\sigma} \log p_{\text{data}}(\mathbf{x}_\sigma)$.
- $\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma)$ is trained to **denoise** the noised samples $\mathbf{x}_\sigma$.

Vincent P. A Connection Between Score Matching and Denoising Autoencoders, 2010

Denoising Score Matching

Initial objective:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

Song Y. Generative Modeling by Estimating Gradients of the Data Distribution, blog post, 2021

Denoising Score Matching

Initial objective:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \big\| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) \big\|_2^2 \to \min_{\boldsymbol{\theta}}$$

Noised objective:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \big\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}} \log q(\mathbf{x}_\sigma) \big\|_2^2 \to \min_{\boldsymbol{\theta}}$$

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Denoising Score Matching

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$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \big\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}} \log q(\mathbf{x}_\sigma) \big\|_2^2 \to \min_{\boldsymbol{\theta}}$$

This is equivalent to a denoising task:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \big\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \big\|_2^2 \to \min_{\boldsymbol{\theta}}$$

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Denoising Score Matching

Initial objective:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

Noised objective:

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$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\mathcal{N}(0,\mathbf{I})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x} + \sigma \cdot \boldsymbol{\epsilon}) + \frac{\boldsymbol{\epsilon}}{\sigma} \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

Langevin Dynamics

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l, \quad \boldsymbol{\epsilon}_l \sim \mathcal{N}(0, \mathbf{I}).$$

Song Y. Generative Modeling by Estimating Gradients of the Data Distribution, blog post, 2021

Denoising Score Matching

Theorem

$$
\begin{aligned}
\mathbb{E}_{q(\mathbf{x}_\sigma)} & \underbrace{\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)\|_2^2}_{h(\mathbf{x}_\sigma)} = \\
&= \mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x})\|_2^2 + \text{const}(\boldsymbol{\theta})
\end{aligned}
$$

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Denoising Score Matching

Theorem

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \underbrace{\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)\|_2^2}_{h(\mathbf{x}_\sigma)} =$$

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Proof

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} h(\mathbf{x}_\sigma) = \int q(\mathbf{x}_\sigma) h(\mathbf{x}_\sigma) d\mathbf{x}_\sigma$$

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Denoising Score Matching

Theorem

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \underbrace{\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)\|_2^2}_{h(\mathbf{x}_\sigma)} =$$

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Proof

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} h(\mathbf{x}_\sigma) = \int q(\mathbf{x}_\sigma) h(\mathbf{x}_\sigma) d\mathbf{x}_\sigma =$$

$$= \int \left(\int q(\mathbf{x}_\sigma|\mathbf{x}) p_{\text{data}}(\mathbf{x}) d\mathbf{x} \right) h(\mathbf{x}_\sigma) d\mathbf{x}_\sigma = \mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} h(\mathbf{x}_\sigma)$$

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Denoising Score Matching

Theorem

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \underbrace{\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)\|_2^2}_{h(\mathbf{x}_\sigma)} =$$

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Proof

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} h(\mathbf{x}_\sigma) = \int q(\mathbf{x}_\sigma) h(\mathbf{x}_\sigma) d\mathbf{x}_\sigma =$$

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$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)\|_2^2 =$$

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Denoising Score Matching

Theorem

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \underbrace{\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)\|_2^2}_{h(\mathbf{x}_\sigma)} =$$

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Proof

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} h(\mathbf{x}_\sigma) = \int q(\mathbf{x}_\sigma) h(\mathbf{x}_\sigma) d\mathbf{x}_\sigma =$$

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$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)\|_2^2 =$$

$$= \mathbb{E}_{q(\mathbf{x}_\sigma)} \Big[\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma)\|^2 + \underbrace{\|\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)\|_2^2}_{\text{const}(\boldsymbol{\theta})} - 2\mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \Big]$$

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Denoising Score Matching

Theorem

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)\|_2^2 =$$
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Proof (Continued)

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left[\mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma)\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)\right]$$

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Denoising Score Matching

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$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2 =$$
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Proof (Continued)

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left[\mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right] = \int q(\mathbf{x}_\sigma) \left[\mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \frac{\nabla_{\mathbf{x}_\sigma} q(\mathbf{x}_\sigma)}{q(\mathbf{x}_\sigma)} \right] d\mathbf{x}_\sigma$$

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Denoising Score Matching

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Proof (Continued)

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left[\mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma)\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)\right] = \int q(\mathbf{x}_\sigma) \left[\mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma)\frac{\nabla_{\mathbf{x}_\sigma} q(\mathbf{x}_\sigma)}{q(\mathbf{x}_\sigma)}\right] d\mathbf{x}_\sigma =$$
$$= \int \left[\mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma)\nabla_{\mathbf{x}_\sigma} \left(\int q(\mathbf{x}_\sigma|\mathbf{x})p_{\text{data}}(\mathbf{x})d\mathbf{x}\right)\right] d\mathbf{x}_\sigma$$

Vincent P. A Connection Between Score Matching and Denoising Autoencoders, 2010

Denoising Score Matching

Theorem

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)\|_2^2 =$$
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Proof (Continued)

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left[\mathbf{s}^\top_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma)\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)\right] = \int q(\mathbf{x}_\sigma) \left[\mathbf{s}^\top_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma)\frac{\nabla_{\mathbf{x}_\sigma} q(\mathbf{x}_\sigma)}{q(\mathbf{x}_\sigma)}\right] d\mathbf{x}_\sigma =$$
$$= \int \left[\mathbf{s}^\top_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma)\nabla_{\mathbf{x}_\sigma} \left(\int q(\mathbf{x}_\sigma|\mathbf{x})p_{\text{data}}(\mathbf{x})d\mathbf{x}\right)\right] d\mathbf{x}_\sigma =$$
$$= \int\int p_{\text{data}}(\mathbf{x}) \left[\mathbf{s}^\top_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma)\nabla_{\mathbf{x}_\sigma} q(\mathbf{x}_\sigma|\mathbf{x})\right] d\mathbf{x}_\sigma d\mathbf{x}$$

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Denoising Score Matching

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Proof (Continued)

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left[\mathbf{s}^\top_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma)\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)\right] = \int q(\mathbf{x}_\sigma) \left[\mathbf{s}^\top_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma)\frac{\nabla_{\mathbf{x}_\sigma} q(\mathbf{x}_\sigma)}{q(\mathbf{x}_\sigma)}\right] d\mathbf{x}_\sigma =$$
$$= \int \left[\mathbf{s}^\top_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma)\nabla_{\mathbf{x}_\sigma} \left(\int q(\mathbf{x}_\sigma|\mathbf{x}) p_{\text{data}}(\mathbf{x}) d\mathbf{x}\right)\right] d\mathbf{x}_\sigma =$$
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$$= \int\int p_{\text{data}}(\mathbf{x}) q(\mathbf{x}_\sigma|\mathbf{x}) \left[\mathbf{s}^\top_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma)\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x})\right] d\mathbf{x}_\sigma d\mathbf{x}$$

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Denoising Score Matching

Theorem

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Proof (Continued)

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left[\mathbf{s}^\top_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma)\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)\right] = \int q(\mathbf{x}_\sigma) \left[\mathbf{s}^\top_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma)\frac{\nabla_{\mathbf{x}_\sigma} q(\mathbf{x}_\sigma)}{q(\mathbf{x}_\sigma)}\right] d\mathbf{x}_\sigma =$$
$$= \int \left[\mathbf{s}^\top_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma)\nabla_{\mathbf{x}_\sigma}\left(\int q(\mathbf{x}_\sigma|\mathbf{x})p_{\text{data}}(\mathbf{x})d\mathbf{x}\right)\right] d\mathbf{x}_\sigma =$$
$$= \int\int p_{\text{data}}(\mathbf{x}) \left[\mathbf{s}^\top_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma)\nabla_{\mathbf{x}_\sigma} q(\mathbf{x}_\sigma|\mathbf{x})\right] d\mathbf{x}_\sigma d\mathbf{x} =$$
$$= \int\int p_{\text{data}}(\mathbf{x})q(\mathbf{x}_\sigma|\mathbf{x}) \left[\mathbf{s}^\top_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma)\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x})\right] d\mathbf{x}_\sigma d\mathbf{x} =$$
$$= \mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left[\mathbf{s}^\top_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma)\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x})\right]$$

Vincent P. A Connection Between Score Matching and Denoising Autoencoders, 2010

Denoising Score Matching

Theorem

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \underbrace{\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)\|_2^2}_{h(\mathbf{x}_\sigma)} =$$

$$= \mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x})\|_2^2 + \text{const}(\boldsymbol{\theta})$$

Proof (Continued)

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} h(\mathbf{x}_\sigma) = \mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} h(\mathbf{x}_\sigma)$$

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Denoising Score Matching

Theorem

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Proof (Continued)

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$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left[\mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma)\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)\right] = \mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left[\mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma)\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x})\right]$$

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Denoising Score Matching

Theorem

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \underbrace{\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)\|_2^2}_{h(\mathbf{x}_\sigma)} =$$

$$= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x})\|_2^2 + \text{const}(\boldsymbol{\theta})$$

Proof (Continued)

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} h(\mathbf{x}_\sigma) = \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} h(\mathbf{x}_\sigma)$$

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left[\mathbf{s}^\top_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)\right] = \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left[\mathbf{s}^\top_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x})\right]$$

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)\|_2^2 =$$
$$= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left[\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma)\|^2 - 2\mathbf{s}^\top_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x})\right] + \text{const}(\boldsymbol{\theta})$$

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Proof (Continued)

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} h(\mathbf{x}_\sigma) = \mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} h(\mathbf{x}_\sigma)$$

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left[\mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma)\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)\right] = \mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left[\mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma)\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x})\right]$$

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)\|_2^2 =$$
$$= \mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left[\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma)\|^2 - 2\mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma)\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x})\right] + \text{const}(\boldsymbol{\theta}) =$$
$$= \mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x})\|_2^2 + \text{const}(\boldsymbol{\theta})$$

Vincent P. A Connection Between Score Matching and Denoising Autoencoders, 2010

Denoising Score Matching

Denoising Score Matching Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

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Denoising Score Matching Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})}\left\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x})\right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

The optimal MSE predictor is the posterior mean of the regression target:

$$\mathbf{s}_{\boldsymbol{\theta}^*,\sigma}(\mathbf{x}_\sigma) = \mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)}\left[\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x})\right]$$

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Denoising Score Matching Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})}\left\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma|\mathbf{x})\right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

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Marginal Score via Conditional Scores

$$\nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma) = \frac{\nabla_{\mathbf{x}_\sigma}\int q(\mathbf{x}_\sigma|\mathbf{x})p_{\text{data}}(\mathbf{x})d\mathbf{x}}{q(\mathbf{x}_\sigma)}$$

Vincent P. A Connection Between Score Matching and Denoising Autoencoders, 2010

Denoising Score Matching

Denoising Score Matching Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})}\left\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma|\mathbf{x})\right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

The optimal MSE predictor is the posterior mean of the regression target:

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Marginal Score via Conditional Scores

$$\nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma) = \frac{\nabla_{\mathbf{x}_\sigma}\int q(\mathbf{x}_\sigma|\mathbf{x})p_{\text{data}}(\mathbf{x})d\mathbf{x}}{q(\mathbf{x}_\sigma)} =$$

$$= \int \nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma|\mathbf{x}) \cdot \frac{q(\mathbf{x}_\sigma|\mathbf{x})p_{\text{data}}(\mathbf{x})}{q(\mathbf{x}_\sigma)}d\mathbf{x}$$

Vincent P. A Connection Between Score Matching and Denoising Autoencoders, 2010

Denoising Score Matching

Denoising Score Matching Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

The optimal MSE predictor is the posterior mean of the regression target:

$$\mathbf{s}_{\boldsymbol{\theta}^*,\sigma}(\mathbf{x}_\sigma) = \mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)} \left[\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x})\right]$$

Marginal Score via Conditional Scores

$$\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) = \frac{\nabla_{\mathbf{x}_\sigma} \int q(\mathbf{x}_\sigma|\mathbf{x}) p_{\text{data}}(\mathbf{x}) d\mathbf{x}}{q(\mathbf{x}_\sigma)} =$$

$$= \int \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \cdot \frac{q(\mathbf{x}_\sigma|\mathbf{x}) p_{\text{data}}(\mathbf{x})}{q(\mathbf{x}_\sigma)} d\mathbf{x} = \mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)} \left[\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x})\right]$$

Vincent P. A Connection Between Score Matching and Denoising Autoencoders, 2010

Denoising Score Matching

Denoising Score Matching Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})}\left\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma|\mathbf{x})\right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

The optimal MSE predictor is the posterior mean of the regression target:

$$\mathbf{s}_{\boldsymbol{\theta}^*,\sigma}(\mathbf{x}_\sigma) = \mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)}\left[\nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma|\mathbf{x})\right]$$

Marginal Score via Conditional Scores

$$\nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma) = \frac{\nabla_{\mathbf{x}_\sigma}\int q(\mathbf{x}_\sigma|\mathbf{x})p_{\text{data}}(\mathbf{x})d\mathbf{x}}{q(\mathbf{x}_\sigma)} =$$

$$= \int \nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma|\mathbf{x})\cdot\frac{q(\mathbf{x}_\sigma|\mathbf{x})p_{\text{data}}(\mathbf{x})}{q(\mathbf{x}_\sigma)}d\mathbf{x} = \mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)}\left[\nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma|\mathbf{x})\right]$$

Therefore: $\mathbf{s}_{\boldsymbol{\theta}^*,\sigma}(\mathbf{x}_\sigma) = \nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma)$ – the optimal model learns the true **marginal** score!

Vincent P. A Connection Between Score Matching and Denoising Autoencoders, 2010

Denoising Score Matching

Original objective:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})\|_2^2 \rightarrow \min_{\boldsymbol{\theta}}$$

Song Y. Generative Modeling by Estimating Gradients of the Data Distribution, blog post, 2021

Denoising Score Matching

Original objective:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})\|_2^2 \to \min_{\boldsymbol{\theta}}$$

Noisy objective:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}} \log q(\mathbf{x}_\sigma)\|_2^2 \to \min_{\boldsymbol{\theta}}$$

Song Y. Generative Modeling by Estimating Gradients of the Data Distribution, blog post, 2021

Denoising Score Matching

Original objective:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})\|_2^2 \to \min_{\boldsymbol{\theta}}$$

Noisy objective:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}} \log q(\mathbf{x}_\sigma)\|_2^2 \to \min_{\boldsymbol{\theta}}$$

This is equivalent to a denoising task:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x})\|_2^2 \to \min_{\boldsymbol{\theta}}$$

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Original objective:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})\|_2^2 \to \min_{\boldsymbol{\theta}}$$

Noisy objective:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}} \log q(\mathbf{x}_\sigma)\|_2^2 \to \min_{\boldsymbol{\theta}}$$

This is equivalent to a denoising task:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x})\|_2^2 \to \min_{\boldsymbol{\theta}}$$

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{\mathcal{N}(0,\mathbf{I})} \left\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x} + \sigma\boldsymbol{\epsilon}) + \frac{\boldsymbol{\epsilon}}{\sigma}\right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

Song Y. Generative Modeling by Estimating Gradients of the Data Distribution, blog post, 2021

Denoising Score Matching

Original objective:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})\|_2^2 \to \min_{\boldsymbol{\theta}}$$

Noisy objective:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}} \log q(\mathbf{x}_\sigma)\|_2^2 \to \min_{\boldsymbol{\theta}}$$

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Langevin Dynamics

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l, \quad \boldsymbol{\epsilon}_l \sim \mathcal{N}(0, \mathbf{I})$$

Song Y. Generative Modeling by Estimating Gradients of the Data Distribution, blog post, 2021

Outline

1. Langevin Dynamics

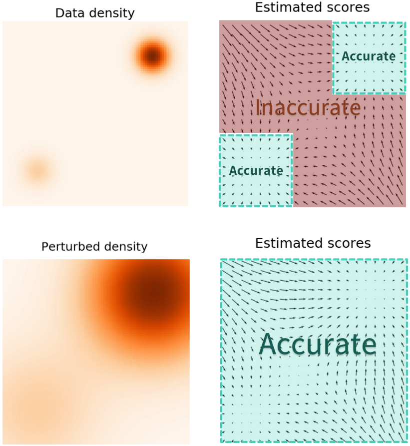
2. Score Matching
   - Denoising Score Matching
   - Noise-Conditioned Score Network (NCSN)

Denoising Score Matching

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{\mathcal{N}(0,\mathbf{I})}\left\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}+\sigma\boldsymbol{\epsilon})+\frac{\boldsymbol{\epsilon}}{\sigma}\right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2}\cdot\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_l) + \sqrt{\eta}\cdot\boldsymbol{\epsilon}_l$$

- For **small** σ, $\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x})$ becomes inaccurate and Langevin dynamics fails to traverse modes.
- For **large** σ, robustness in low-density regions is achieved, but the model learns a distribution that is overly corrupted.



Song Y. Generative Modeling by Estimating Gradients of the Data Distribution, blog post, 2021

Noise-Conditioned Score Network (NCSN)

- Specify a sequence of noise levels: $\sigma_1 < \sigma_2 < \cdots < \sigma_T$.
- Perturb each data point with different noise levels: $\mathbf{x}_t = \mathbf{x} + \sigma_t \epsilon$, so $\mathbf{x}_t \sim q(\mathbf{x}_t)$.
- Choose σ_1, σ_T such that:

$$q(\mathbf{x}_1) \approx p_{\text{data}}(\mathbf{x}), \quad q(\mathbf{x}_T) \approx \mathcal{N}(0, \sigma_T^2 \mathbf{I})$$

Song Y. et al. Generative Modeling by Estimating Gradients of the Data Distribution, 2019

Noise-Conditioned Score Network (NCSN)

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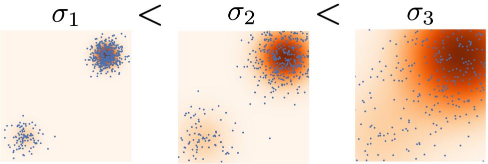

Song Y. et al. Generative Modeling by Estimating Gradients of the Data Distribution, 2019

Noise-Conditioned Score Network (NCSN)

Train the denoising score function $\mathbf{s}_{\boldsymbol{\theta},\sigma_t}(\mathbf{x}_t)$ for each noise level σ_t using a unified weighted objective:

$$\sum_{t=1}^{T} \sigma_t^2 \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma_t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

Song Y. et al. Generative Modeling by Estimating Gradients of the Data Distribution, 2019

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Here, $\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}) = -\frac{\mathbf{x}_t - \mathbf{x}}{\sigma_t^2} = -\frac{\boldsymbol{\epsilon}}{\sigma_t}$

Song Y. et al. Generative Modeling by Estimating Gradients of the Data Distribution, 2019

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Here, $\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}) = -\frac{\mathbf{x}_t - \mathbf{x}}{\sigma_t^2} = -\frac{\boldsymbol{\epsilon}}{\sigma_t}$

Training

1. Sample $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$, $t \sim U\{1, T\}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$.
2. Compute noisy image $\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \boldsymbol{\epsilon}$.
3. Compute loss $\mathcal{L} = \sigma_t^2 \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma_t}(\mathbf{x}_t) + \frac{\boldsymbol{\epsilon}}{\sigma_t} \right\|^2$.
4. Update $\boldsymbol{\theta}$ via stochastic gradient descent.

Song Y. et al. Generative Modeling by Estimating Gradients of the Data Distribution, 2019

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Train the denoising score function $\mathbf{s}_{\boldsymbol{\theta},\sigma_t}(\mathbf{x}_t)$ for each noise level σ_t using a unified weighted objective:

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Training

1. Sample $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$, $t \sim U\{1, T\}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$.
2. Compute noisy image $\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \boldsymbol{\epsilon}$.
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4. Update $\boldsymbol{\theta}$ via stochastic gradient descent.

How do we sample from such a model?

Song Y. et al. Generative Modeling by Estimating Gradients of the Data Distribution, 2019

Noise-Conditioned Score Network (NCSN)

Sampling (Annealed Langevin Dynamics)

1. Sample $\mathbf{x}_T^0 \sim \mathcal{N}(0, \sigma_T^2 \mathbf{I}) \approx q(\mathbf{x}_T)$.
2. Update $\mathbf{x}_t^l$ via L steps of Langevin dynamics at each noise level σ_t:
$$\mathbf{x}_t^l = \mathbf{x}_t^{l-1} + \frac{\eta_t}{2} \mathbf{s}_{\boldsymbol{\theta}, \sigma_t}(\mathbf{x}_t^{l-1}) + \sqrt{\eta_t} \boldsymbol{\epsilon}_t^l.$$
3. Update $\mathbf{x}_{t-1}^0 := \mathbf{x}_t^L$ and proceed to the next noise level σ_{t-1}.

Song Y. et al. Improved Techniques for Training Score-Based Generative Models, 2020

Noise-Conditioned Score Network (NCSN)

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3. Update $\mathbf{x}_{t-1}^0 := \mathbf{x}_t^L$ and proceed to the next noise level σ_{t-1}.

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Summary

- Langevin dynamics enables sampling from unnormalized densities (e.g. EBMs) using the gradient of the log-density $\nabla_{\mathbf{x}} \log p_{\boldsymbol{\theta}}(\mathbf{x})$.
- Score matching proposes minimizing Fisher divergence to estimate the score function.
- Denoising score matching minimizes the Fisher divergence on corrupted samples, making the divergence estimable via sampling.
- The Noise-Conditioned Score Network leverages a range of noise levels and annealed Langevin dynamics to learn the score function and enable sampling.